18.01.2021

CEN 301 Data Management and File Structures Final Exam

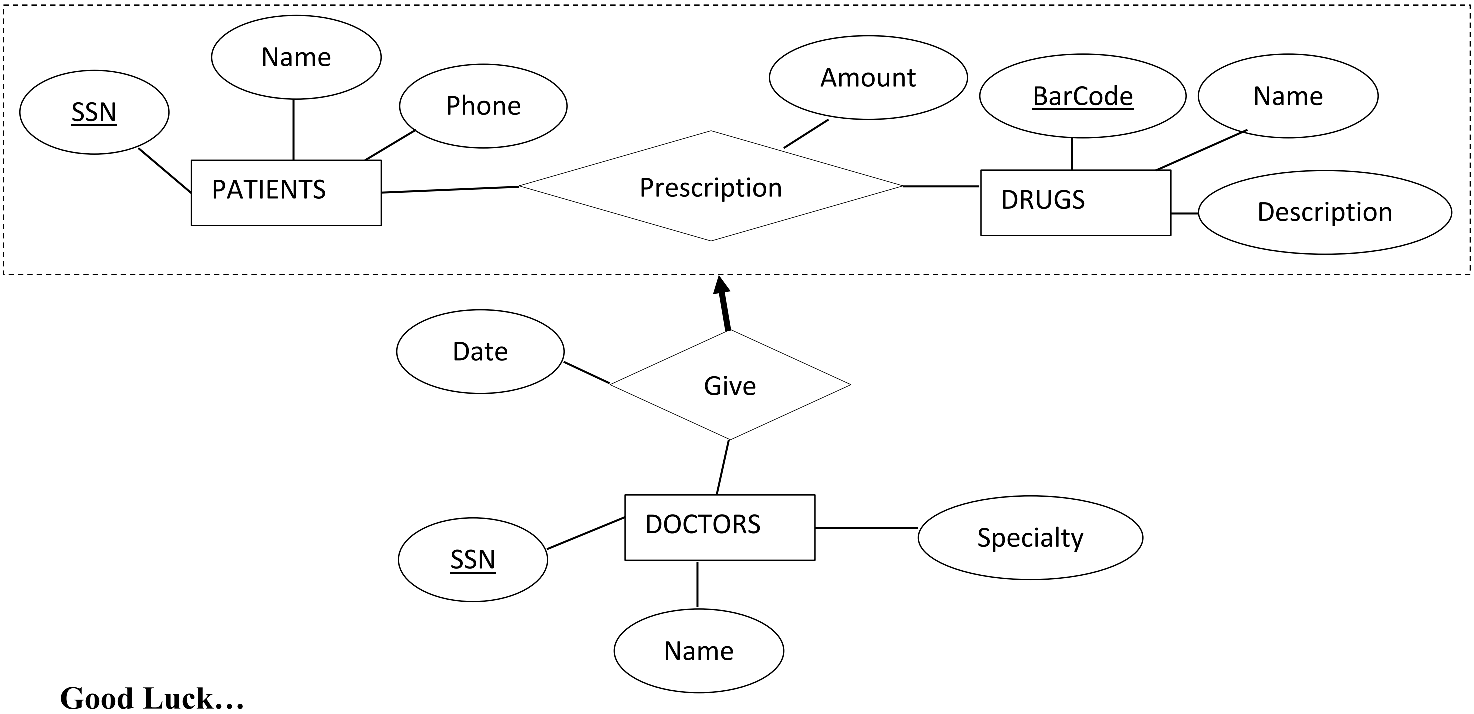
QUESTIONS

(*There are 3 questions in this exam, you should submit your document(s) until 14:40*)

1. **(35 points)** Consider a disk with a block size of 2000 bytes, block transfer time of 0.5 milliseconds, average rotational delay of 7 milliseconds, and average seek time of 10 milliseconds. Suppose that a file containing 150000 records of 300 bytes each is to be stored on such disk and that no record is allowed to span two blocks.
   a) Compute number of passes, number of disk accesses, and time (in milliseconds) required to sort this file using **general external merge sort (without any improvement)** algorithm assuming that 11 buffer pages are available in the memory.
   b) Assume that an **unclustered B+tree index using Alternative 2** is constructed on the file. In this index, you can assume that the key field and address pair occupy 100 bytes for each record, and each leaf node is 60% full. Only the leaf nodes and the file are stored on disk. Compute time required for sorting the file by using the unclustered B+tree index.

2. **(35 points)** A project management database contains information about projects (identified by project-id, and they have title, description, start date, end date, and budget), and employees (identified by SSN, each employee has name, surname, specialty, and phone number). A project has any number of project packets but each packet (identified by packet id, title, duration) must belong to an exactly one project. Project packets are assigned to employees. For each packet, at least one employee is assigned to, and employee can be assigned to any number of packets for a particular duration. Some of the employees are managers and they supervise employees who are assigned to project packets. For each assignment of employee to any project packet, exactly one manager must supervise. However, a manager can supervise any number of assignments. Draw an ER diagram that describes the given information (assuming no further constraints hold).

3. **(30 points)** Write SQL commands to create the database schema that describes best the below E/R diagram. When a patient is deleted, all prescriptions belong to him/her must also be deleted. Your commands must include all the tables, attributes, primary keys, and foreign keys constraints.



Good Luck…